The Case for Voluntary Disclosure

Strategic Feature Revelation in Regression and Classification

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Strategic Learning with Heterogeneous Preferences

Roadmap

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1. Motivation & Core Question
2. Problem Formulation
3. Game-Theoretic Model

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5. Sufficiency & Necessity Conditions
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Motivation

The Disclosure Dilemma

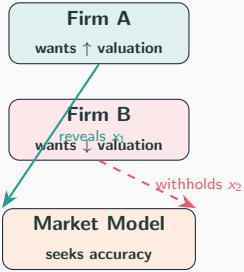
Consider a predictive model (e.g., firm valuation):

- **Mandatory:** All features disclosed $\Rightarrow$ baseline accuracy
- **Voluntary:** Participants *choose* what to reveal

Key insight: Participants have **private preferred outcomes** and will strategically withhold unfavorable features.

Surprising Twist

Selective disclosure can sometimes **help** the model by filtering out noisy signals!



Why Heterogeneous Preferences Matter

Uniform Preferences (Easy Case)

If *all* firms want overvaluation $\Rightarrow$ assign a conservative default.

Strategic withholding is trivially discouraged.

Heterogeneous Preferences (Hard Case)

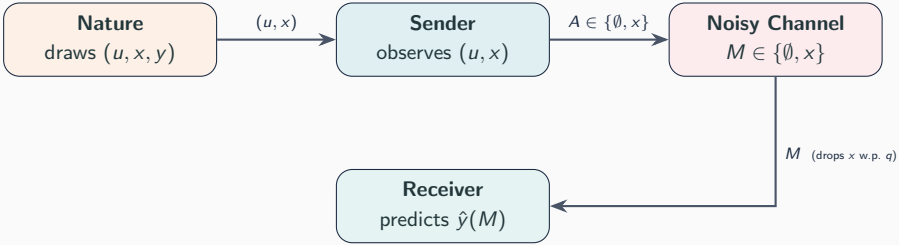
Firms have *different* preferred directions $\Rightarrow$ no single default works.

This is the realistic & challenging setting we study.

Our Contribution: Formal conditions characterizing *exactly when* voluntary disclosure benefits the predictive model—for **regression**, **classification**, and **online learning**.

Problem Formulation

Game-Theoretic Model



Sender loss: $\ell(u, \hat{y})$ | Receiver loss: $\ell(y, \hat{y})$

Sender Strategy

$\mathcal{S} = \{A : \mathcal{Y} \times \mathcal{X} \mapsto \{\emptyset, x\}\}$

Disclose full features x or withhold $\emptyset$

Receiver Strategy

$\mathcal{R} = \{\hat{y} : \{\emptyset, \mathcal{X}\} \mapsto \mathcal{Y}\}$

Predict based on received message M

Stackelberg Equilibrium & Strategic Advantage

We model the interaction as a **screening game** (Stackelberg):

1. **Receiver commits** to a strategy $\hat{y} \in \mathcal{R}$
2. **Sender best-responds:** $A(u, x) = \arg\min_{a \in \{\emptyset, x\}} \mathbb{E}[\ell(u, \hat{y}(M)) \mid A = a]$
3. **Outcome realized** and losses incurred

Strategic Risk (Voluntary)

$$R_S := \min_{\hat{y} \in \mathcal{R}} \mathbb{E}[\ell(Y, \hat{y}(M))]$$

Vanilla Risk (Mandatory)

$$R_V := \mathbb{E}[\ell(Y, \hat{y}_V(M))]$$

Definition: Strategic Advantage

A **strategic advantage** exists if $R_S < R_V$ — voluntary disclosure *strictly outperforms* mandatory.

Regression Results

Regression Setup

- Outcome space: $\mathcal{Y} = \mathbb{R}$, Loss: $\ell(y, \hat{y}) = (y - \hat{y})^2$ (MSE)
- All variables (U, X, Y) have **zero mean**
- Linear receiver: $\hat{y}(\emptyset) = 0$, $\hat{y}(x) = b^\top x$

Vanilla Risk (Mandatory)

$$R_V^\star(q) = \mathrm{Var}(Y) - (1-q)\,\mathrm{Var}(b_Y^\top X)$$

where $b_Y = \mathrm{Var}(X)^{-1}\,\mathrm{Cov}(X, Y)$
is the OLS weight vector.

Strategic Risk (Voluntary)

$$R_S(b, q) = \mathrm{Var}(Y) - (1-q)\mathbb{E}[G \cdot \mathbb{I}\{F > 0\}]$$

Key scalar fields governing strategic behavior:

$F(x, u) := (2u - b^\top x)\, b^\top x$ — Sender's decision rule

$G(x, y) := (2y - b^\top x)\, b^\top x$ — Receiver's gain function

Sufficiency Condition for Regression

Theorem (Sufficiency)

A **strategic advantage** exists if $\exists\, b \in \mathbb{R}^d$ such that:

$$\|U - Y\|_2 < \frac{\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X)}{4\sqrt{\mathrm{Var}(b_Y^\top X)}}$$

LHS: $\|U - Y\|_2$

Measures *preference misalignment* between sender's preferred value U and truth Y

RHS: Feature informativeness

Driven by $\mathbb{E}[|G|]$, the expected gain from observing features, minus sub-optimality penalty

Strategic advantage $\iff$ **preferences are sufficiently aligned** relative to feature informativeness

Necessary Condition for Regression

Theorem (Necessity)

A **necessary condition** for a strategic advantage to exist:

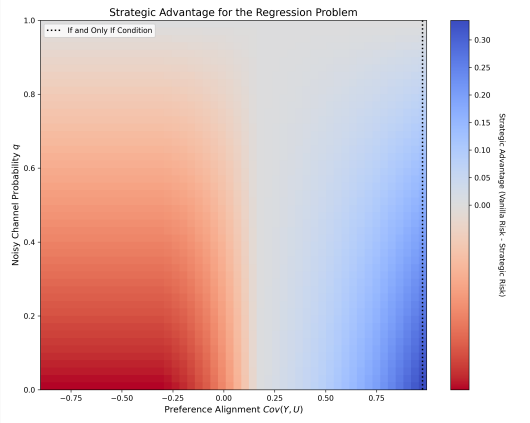
$$\max_b \left(\mathbb{E}\big[|G(X,Y)|\big] - \mathrm{Var}(b_Y^\top X) - \mathrm{Var}((b - b_Y)^\top X) \right) > 0$$

Key Remarks

- The necessary condition ensures the sufficiency bound is **non-vacuous**
- ⋆ **Both conditions are independent of channel noise q!**
  q controls the *magnitude* of the advantage, not its *existence*
- If conditions hold at $b = b_Y$ (vanilla OLS weights), they hold for optimal b

Takeaway: Whether voluntary disclosure helps is a *structural* property of the distribution—not a consequence of the channel.

Regression: Numerical Validation



Setup:

- $(X, Y, U) \sim \mathcal{N}(0, \Sigma)$
- Unit marginal variances
- $\mathrm{Cov}(X, Y) = \mathrm{Cov}(X, U) = 0.2$

Parameter sweeps:

- $q \in [0, 1]$ (channel noise)
- $\mathrm{Cov}(Y, U) \in [-1, 1]$ (preference alignment)

Observation

The **dotted contour** (from our analytical sufficiency bound) reliably delineates the true region of strategic advantage.

Gaussian Specialization

Gaussian Specialization

When $(U, X, Y) \sim \mathcal{N}(0, \Sigma)$ are jointly Gaussian, the sufficiency condition admits a **closed-form** expression.

Theorem (Gaussian Sufficiency)

With receiver predictor $b = 2\Sigma_{XX}^{-1}\Sigma_{XU}$, a strategic advantage exists if:

$$2\left(b_Y^\top \Sigma_{XU} - \sigma_Z^2\right) + \frac{8\kappa\,\sigma_Z}{s\sqrt{2\pi}}\, e^{-\alpha^2/2} \;>\; \Sigma_{XY}^\top \Sigma_{XX}^{-1} \Sigma_{XY}$$

Defined Parameters

- $b_Y = \Sigma_{XX}^{-1}\Sigma_{XY}$ (OLS for Y)
- $b_U = \Sigma_{XX}^{-1}\Sigma_{XU}$ (OLS for U)
- $\sigma_Z^2 = \Sigma_{XU}^\top \Sigma_{XX}^{-1}\Sigma_{XU}$

Conditional Parameters

- $\kappa = \mathrm{Cov}(U, Y \mid X)$
- $s^2 = \mathrm{Var}(U \mid X)$
- $\alpha = 0$ (zero mean)

Classification Results

Classification Setup

- Outcome space: $\mathcal{Y} = \{-1, +1\}$, Loss: $\ell(y, \hat{y}) = \mathbb{I}\{\hat{y} \neq y\}$ (0-1 loss)
- Preference: $U \in \{-1, +1\}$ (binary preferred label)
- Assume WLOG: $\mathbb{E}[Y] > 0$ (label $+1$ is the majority class)

Vanilla Risk

$$R_V^\star(q) = q\,\mathbb{P}(Y = -\operatorname{sign}(\mathbb{E}[Y])) + (1-q)\,\mathbb{E}\left[\frac{1 - |\eta(X)|}{2}\right]$$

where $\eta(x) = \mathbb{E}[Y \mid X = x]$.

Strategic Risk

For default option $i \in \{-1, +1\}$:

$$R_{\mathcal{S}_i}^\star(q) = \pi_i\,\mathbb{P}(Y = -i \mid U = i) + \pi_{-i}\left[(1-q)\mathbb{E}\left[\frac{1-|\eta_{-i}|}{2}\right] + q\,\mathbb{P}(Y = -i \mid U = -i)\right]$$

Only **two** possible strategic receiver strategies: default to $+1$ ($\mathcal{S}_1$) or default to -1 ($\mathcal{S}_{-1}$).
The optimal strategy depends on distribution structure and channel noise q.

Classification: Stackelberg Equilibrium

Theorem (Stackelberg Equilibrium & Strategic Advantage)

Assume $\mathbb{E}[Y] > 0$. At max noise ($q = 1$), strategy $\mathcal{S}_1$ is always better. Two scenarios arise:

Case 1: $\mathcal{S}_1$ dominates $\forall q$

When $R^\star_{\mathcal{S}_1}(0) \leq R^\star_{\mathcal{S}_{-1}}(0)$:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_{-1}\mathbb{E}\left[\frac{1-|\eta_{-1}|}{2}\right] - \pi_1\mathbb{P}(Y=-1 \mid U=1)\right] > 0$$

Independent of q!

Analogous to regression.

Case 2: Strategy switch at $q^\star$

$\exists\, q^\star \in (0,1)$. For $q \geq q^\star$: use $\mathcal{S}_1$.

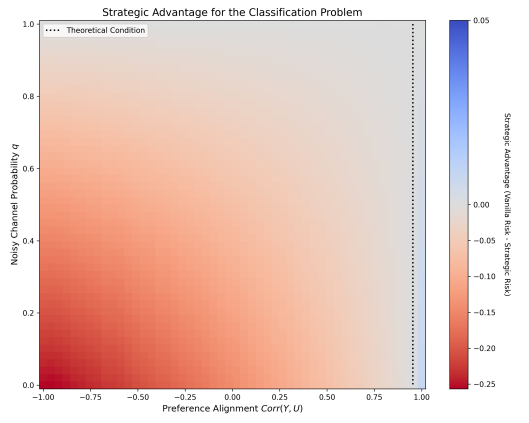
For $q < q^\star$: use $\mathcal{S}_{-1}$ with:

$$(1-q)\left[\mathbb{E}\left[\frac{1-|\eta|}{2}\right] - \pi_1\mathbb{E}\left[\frac{1-|\eta_1|}{2}\right]\right] + q\left[\mathbb{P}(Y=-1) - \pi_1\mathbb{P}(Y=1 \mid U=1)\right] > \pi_{-1}\mathbb{P}(Y=1 \mid U=-1)$$

Depends on q!

Noise affects existence.

Classification: Numerical Validation



Setup:

- $(Y, U) \in \{-1, +1\}^2$
- $P(Y=1) = 0.6,\ P(U=1) = 0.5$
- Features: Gaussian mixture
  $X \mid Y, U \sim Y\mu_Y + U\mu_U + \mathcal{N}(0, 1)$

Regression vs. Classification:

- Regression: advantage region **independent** of q
- Classification: advantage region **can depend** on q (Case 2)

Validation

Analytical boundary matches the empirical advantage region precisely.

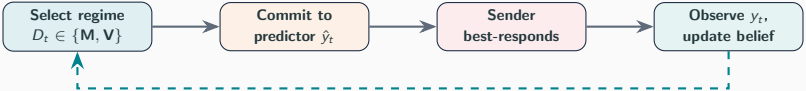
Online Learning

Dynamic Setting: Unknown Environment

The Challenge

All previous results assumed the environment (joint distribution of (X, Y, U)) is **known**.
What if the receiver must **learn** it?

Protocol: Over T rounds, at each round t:



Objective: Minimize cumulative regret:

$$\mathcal{R}_T = \sum_{t=1}^{T} \ell\big(y_t, \hat{y}_t(M_t)\big) - T \cdot \min\{R_V^\star, R_S^\star\}$$

Bayesian Adaptive Disclosure Protocol

Algorithm Overview

1. **Initialize** prior $p_1(\theta)$ over parameter space Θ
2. **Each round** t**:**
   - Evaluate expected risk under belief p_t for both regimes
   - Select $(D_t, \hat{y}_t)$ minimizing expected loss
   - Observe outcome, update posterior via Bayes' rule:

$$p_{t+1}(\theta) \propto p_t(\theta) \cdot \mathbb{P}(M_t, y_t \mid \theta, \hat{y}_t)$$

Regret Guarantee

Under compact Θ and standard regularity:

$$\boxed{\mathbb{E}[\mathcal{R}_T] \leq C\sqrt{T}}$$

Sublinear regret $\Rightarrow$ algorithm **converges** to the optimal disclosure regime.

Regret Decomposition

- **Estimation:** $O(\sqrt{T})$
  standard online regression
- **Regime Selection:** $O(\log T)$
  posterior concentration

Conclusion

Summary of Contributions

Regression

Sufficiency & necessity conditions for strategic advantage

Independent of q

Gaussian closed-form

→

Classification

Full Stackelberg equilibrium characterization

Two-case structure

q-dependent threshold

→

Online Learning

Bayesian adaptive disclosure protocol

$O(\sqrt{T})$ regret

Converges to optimal

Unifying Theme: *Preference alignment* between sender and ground truth is the key structural property that determines whether voluntary disclosure is beneficial.

Limitations & Future Directions

Limitations

- Sufficiency bounds can be **conservative** for some distribution families
- Restricted to **linear** receiver strategies in regression
- Online learning assumes **parametric** environment

Future Directions

- **Non-linear predictors:** kernels, neural networks
- **Multi-sender** competitive environments
- **Partial disclosure:** reveal subsets of features rather than all-or-nothing
- **Tighter bounds:** closing the gap between sufficiency and necessity

Open Question

Can we design *incentive-compatible* mechanisms that encourage truthful disclosure while maintaining the performance benefits of voluntary revelation?

Thank You!

Questions & Discussion

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Key Message

Voluntary disclosure can outperform mandatory disclosure when participant preferences are sufficiently aligned with ground truth.